

What is the difference between "dimension of a vector" and "dimension of a subspace"?

Answer:

The two terms sound similar, but they describe fundamentally different things: one describes a property of a single vector, while the other describes a property of a set of vectors.

Here's a breakdown of the difference.

Strictly speaking, in formal mathematics, the term "**dimension of a vector**" is not used.

A single vector does not have a dimension.

When we use this phrase informally in our course MA1513, we always mean **the number of components in the vector**, which tells you the dimension of the larger space the vector lives in.

- **What it describes:** The size or number of entries in a single vector.
- **Example:** The vector $\mathbf{v} = [4, -1, 8]$ has 3 components. You might informally say it's a "3-dimensional vector" because it belongs to the vector space $\mathbf{R}^3$.
- **How to find it:** Simply count the number of components.

"**Dimension of a Subspace**" is a formal and very important concept in linear algebra.

The dimension of a subspace is a measure of its "size" or "degrees of freedom."

- **What it describes:** The number of linearly independent vectors required to span the entire subspace. This number is unique for any given subspace.
- **Formal Definition:** The dimension of a subspace is the **number of vectors in any basis** for that subspace. (A **basis** is a set of vectors that is both **linearly independent** and **spans** the subspace.)
- **How to find it:** You must find a basis for the subspace and then count how many vectors are in that basis.
- **Analogy:** Think of giving directions.
 - If your subspace is a **line** through the origin in $\mathbf{R}^3$, you only need one direction vector and a distance (e.g., "go 5 units in *this* direction"). Since you only need **one** direction vector to define the entire line, the dimension of the line is **1**.
 - If your subspace is a **plane** through the origin in $\mathbf{R}^3$, you need two independent directions (e.g., "go 2 units in *this* direction, then 3 units in *that* direction"). Since you need **two** basis vectors to reach any point on the plane, the dimension of the plane is **2**.

Summary Table

Attribute	"Dimension of a Vector" (Informal)	"Dimension of a Subspace" (Formal)
What it applies to	A single vector.	An entire set of vectors (the subspace).
What it measures	The number of components in the vector.	The number of independent vectors needed to "build" the entire subspace.
How it's determined	By counting the components.	By finding a basis and counting its vectors.

Scenario 1

A line in 2D plane that passes through the origin is a subspace in $\mathbf{R}^2$.

For example, the line L with equation $y = 3x$ contains the vector $(1, 3)$ and all scalar multiples of $(1,3)$. So $L = \text{span}\{ (1,3) \}$. This is a subspace of $\mathbf{R}^2$ with a single basis vector $(1,3)$. Since there is only 1 vector in the basis, the dimension of L is 1.

Note that every vector in L comes from $\mathbf{R}^2$, and so each vector has "dimension" 2 which is different from the dimension of L.

Scenario 2

A plane in 3D space that contains the origin is a subspace in $\mathbf{R}^3$.

For example, the plane P with equation $x + y - 2z = 0$ contains the vectors $(1, 1, 1)$ and $(2, 0, 1)$ and all linear combinations of $(1, 1, 1)$ and $(2, 0, 1)$. So $L = \text{span}\{ (1,1,1), (2,0,1) \}$. This is a subspace of $\mathbf{R}^3$ with a two basis vectors $(1,1,1)$ and $(2,0,1)$. Since there are two vectors in the basis, the dimension of P is 2.

Note that every vector in P comes from $\mathbf{R}^3$, and so each vector has "dimension" 3, which is different from the dimension of L.

Scenario 3

The 4-dimensional space $\mathbf{R}^4$ contains all vectors of the form (a, b, c, d) . Every vector in $\mathbf{R}^4$ has "dimension" 4.

$\mathbf{R}^4$ has a standard basis $S = \{ (1,0,0,0), (0,1,0,0), (0,0,1,0), (0,0,0,1) \}$.

- Check that every vector (a, b, c, d) in $\mathbf{R}^4$ can be written as a linear combination of the four vectors in S: $(a,b,c,d) = a(1,0,0,0)+b(0,1,0,0)+c(0,0,1,0)+d(0,0,0,1)$; and conversely, every linear combination of the four vectors is again a vector of $\mathbf{R}^4$.
- So $\mathbf{R}^4 = \text{span}(S)$
- The four vectors in S can be easily checked to be linearly independent.
- So S forms a basis for $\mathbf{R}^4$.

By counting the number of vectors in the basis, the dimension of $\mathbf{R}^4$ is 4.
 More generally, a basis for $\mathbf{R}^n$ has n vectors, and the dimension of $\mathbf{R}^n$ is n .

Scenario 4

$\text{Span}\{(1,0,0,0), (0,1,0,0), (1,1,0,0)\}$ is the set of all linear combinations of the three vectors $(1,0,0,0), (0,1,0,0), (1,1,0,0)$ coming from $\mathbf{R}^4$.

Let us denote this set by W . i.e. $W = \text{Span}\{(1,0,0,0), (0,1,0,0), (1,1,0,0)\}$.

By default, W is a subspace of $\mathbf{R}^4$, being a linear span of a set of 4-dimensional vectors. To find a basis for W , we look at the set $\{(1,0,0,0), (0,1,0,0), (1,1,0,0)\}$. By inspection, the third vector $(1,1,0,0)$ is the sum of the first two vectors, and hence is a “redundant” vector. the remaining two vectors $(1,0,0,0)$ and $(0,1,0,0)$ are not scalar multiple of each other, so they are linearly independent. Therefore, we can take $(1,0,0,0)$ and $(0,1,0,0)$ as a basis for W . This gives the dimension of W to be 2.

In general, given $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ where $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ are n -dimensional vectors in $\mathbf{R}^n$. While $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is a subspace of $\mathbf{R}^n$, its dimension need not be k or n . We need to find a basis for $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$. It may not always be easy to do this by inspection. There are other systematic approach such as the “column space” method to identify and remove redundant vectors from $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ to get a basis for the subspace.

Here is an example to illustrate the “column space method”:

Let's define $V = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4, \mathbf{v}_5\}$ with the following vectors:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \mathbf{v}_4 = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}, \mathbf{v}_5 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Form a Matrix $\mathbf{A}$ with these Vectors as Columns

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 & 2 & 0 \\ 1 & 1 & 2 & 2 & 0 \\ 0 & 2 & 2 & 0 & 1 \end{bmatrix}$$

The RREF is given by: $\begin{bmatrix} 1 & 0 & 1 & 2 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

Based on our RREF:

- Pivot columns are 1, 2, and 5.
- Non-pivot (redundant) columns are 3 and 4.

Therefore, the **redundant vectors** are $\mathbf{v}_3$ and $\mathbf{v}_4$ and a **basis for V** is $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_5\}$.

The dimension of the subspace V is 3.

Scenario 5

Solution space of the homogeneous system

$$\begin{aligned} x_1 + 2x_2 - x_3 + 3x_4 &= 1 \\ 2x_1 + 4x_2 + x_3 + 0x_4 &= 2 \end{aligned}$$

Let us denote this solution space by U .

Using GE/GJE, we get the general solution: $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = s \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ 2 \\ 1 \end{bmatrix}$ after separating

parameters. From here, we see that all the solutions can be expressed as linear

combinations of the two vectors $\mathbf{v}_1 = \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} -1 \\ 0 \\ 2 \\ 1 \end{bmatrix}$. Since $\mathbf{v}_1$ and $\mathbf{v}_2$ are linearly

independent, they form a basis $\{\mathbf{v}_1, \mathbf{v}_2\}$ for this solution space U. Hence the dimension of U is given by the number of basis vectors, which is 2.

- Each vector in that basis is a "**4-dimensional vector**" (informal term) in $\mathbf{R}^4$ because the vector has 4 components corresponding to the 4 variables of the system.
- The **dimension of the solution space** is **2**, because its basis $\{\mathbf{v}_1, \mathbf{v}_2\}$ contains two vectors. This number actually comes from the number of parameters in the general solution of the system.